



Software Requirements Specification

According to ISO/IEC/IEEE 16326 Guidelines

Course: Software Construction

Submitted by: Group 1 (Software Construction Class)

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Table 2: Task Table

Signature Page

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Change History

Version	Author	Date	Change Description
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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to outline the requirements for the development of GiftCart software. This document will serve as a foundational guideline for the project's scope, detailing all necessary functionalities, constraints, and any non-functional requirements. This SRS serves as a base of contract between the customer and the supplier.

The SRS is intended for the following audience:

- a. **Development Team:** To provide a clear understanding of the requirements that need to be implemented.
- b. **Customer:** To validate whether the requirements outlined are the ones which he needed.
- c. **Stakeholders:** To validate whether the requirements align with project objectives and goals.
- d. **Managers:** To ensure all requirements are properly documented for resource allocation and scheduling.
- e. **Testers:** To facilitate test planning and to ensure all requirements are met and validated.

1.2 Scope

1.2.1. Name

The name of the project to be developed is **GiftCart**.

1.2.2. Main Function

- a. **Registry Management:** Allows users to create, view, and manage event-specific gift registries, including item selection and updates.
- b. **User Interaction:** Enables friends to browse registries, select gifts, and make purchases without revealing others' selections.
- c. **Inventory Control:** Automatically updates item availability based on gift selections, ensuring accurate stock management.
- d. **Payment Integration:** Facilitates secure credit card payments for selected gifts within the system.
- e. **Deadline Management:** Manages event deadlines for gift selection, deactivating registries as needed.

1.2.3. Objectives

- a. **Privacy and Dignity:** Ensure that friends selecting gifts cannot see what others have chosen, maintaining privacy and dignity during the gift selection process.
- b. **Unique Gift Selection:** Ensure that the host receives only the specific gifts they requested in the required quantities, preventing duplicate selections and ensuring the host gets exactly what they need.

- c. **Accessibility:** Provide an intuitive and accessible platform for both hosts (event organizers) and guests (friends), allowing easy access to registries and gift selection from any device even from remote areas.
- d. **Deadline and Event Coordination:** Manage event deadlines effectively, including coordination of gift selection and payment deadlines, to ensure timely gift purchases and event preparation.
- e. **Seamless Payment Integration:** Implement secure, seamless credit card payment processing, ensuring guests can complete transactions efficiently and without difficulty.
- f. **Receipt Generation:** Automatically generate and distribute receipts for both guests and the shop, confirming payment completion and allowing for easy tracking of transactions as a proof of gift.
- g. **Inventory Management:** Keep real-time track of available and selected items, updating the shop's inventory and the gift registry to reflect changes as gifts are purchased.
- h. **Registry Deactivation:** Automatically deactivate or delete the registry once the event deadline has passed or all items have been selected and paid for.

1.3 Definition, Acronyms and Abbreviations

1.3.1. Definitions

- 1. **Gift Registry:** A list of items that a person compiles, typically for an event such as a wedding or birthday, allowing others to purchase gifts from the list.
- 2. **Host (or Registered Person):** The person who creates the gift registry for their event.
- 3. **Guest (or Friend):** A person who accesses the gift registry to select and purchase gifts for the host's event.
- 4. **Admin (or Shop Owner):** The person who manages the shop catalogue.
- 5. **Event Deadline:** A date set by the host and the shop, after which the gift registry is disabled, and purchases are no longer allowed.
- 6. **Java:** A high-level, object-oriented programming language used for developing gift registry software. *Reference: Oracle Java Documentation.*
- 7. **Project Libre:** free, open-source project management software used to create the schedule and manage tasks for this project.
- 8. **Frontend:** The part of the software system that interacts directly with users, including the graphical user interface (GUI) where hosts and guests can interact with the registry.
- 9. **Backend:** The server-side part of the software that handles data processing, business logic, database management, and communication with the frontend.
- 10. **System:** The GiftCart system to be developed.

1.4 References

- 1. **ISO/IEC/IEEE 16326:2019**
Systems and Software Engineering — Life Cycle Processes — Project Management
Description: Provides the guidelines for structuring and content of the project plan.
- 2. **Java Documentation**
Oracle's official documentation on Java, used as the programming language for developing the

system.

Source: <https://docs.oracle.com/javase/>

3. **Project Management Guidelines**

Course materials provided by the professor, covering project management and planning principles, feedback processes, and the use of Project Libre.

1.5 Overview

The rest of the document contains an abstract and detailed description of the system requirements for technical as well as non-technical audience. It includes functional and non-functional requirements using use cases and system functions, description of functions and sub-functions to be made, user characteristics, assumptions made, and constraint or special requirements specified. The document also contains appendices to support understanding of the requirements using data dictionary, and visual models including case diagram and system sequence diagrams, and domain diagram to support understanding for developer team and technical audience.

The SRS is organized into the following main sections:

1. **Introduction:** Outline's purpose of the document, scope and intended audience as well as relevant references and definitions of terms used.
2. **Overall description:** Outlines product context/perspective to specify its dependency or standalone nature, descriptions of functions, user characteristics to specify the level of users that will use the system and constraints imposed on the development of system and on the product.
3. **Specific Requirements:** Here is outlined the detailed specifications of the requirements without the implementation details, with the aid of use cases, system functions and sub-functions.
4. **Appendix:** This section contains the data dictionary to understand the terms and words of the system specifications used throughout the use cases, system function, use case diagrams, system sequence diagrams and domain diagram.

2. Overall Description

2.1 Product Perspective:

The GiftCart system is standalone system that will be developed in JAVA. Unlike traditional eCommerce platforms the GiftCart system is focused specifically on registry management catering to the unique needs of event-based gifts selection and coordination.

2.2 Product Functions:

1. **Registration:** Users create an account by giving specific details like name, email and password.
2. **Login:** Users authenticate their identity by entering email and password to access their account.
3. **Add items to catalogue:** Admin can add items to catalogue with specific details like name, price and description.
4. **Update items from catalogue:** Admin can modify the details of existing items in the catalogue.
5. **Delete items from catalogue:** Admin can delete items from catalogue.
6. **View catalogue:** Admin and host can see items available in catalogue.
7. **Create event:** Host can create an event providing details like name, date and description.
8. **Add items to registry:** Host can select and add items to registry from catalogue.
9. **Update items in registry:** Host can modify items from gift registry.
10. **Delete items from registry:** Host can delete items from gift registry.
11. **View registry.** Host can see the contents of registry.
12. **Select gift:** Guest can select gift of his/her choice for purchases or reservation.
13. **Update gift:** Guest can change the details or selection of gifts which he have chosen from registry.
14. **Remove gift:** Guest can remove gifts from his cart.
15. **View gift:** Host and Guest can view detailed information about specific gift including its availability.
16. **Make payment:** Guest can complete his purchase by giving payment information.
17. **Backup :** The system performs regular backups for user data, item information and registry details to ensure data integrity.

2.3 User Characteristics:

The GiftCart system is developed for three kind of users that are Admin, Host and Guest.

Admin: The admin responsible for overseeing the entire system, managing inventory, user accounts, and system maintenance. This role requires a high level of technical proficiency including familiarity with systems backend, user account control and troubleshooting basic issues.

Host: The Host is an event organizer who creates and manages gift registries for specific events. They should have the basic understanding of navigating applications creating and updating registries and setting event specific preferences.

Guest: **Guest are** users who browse registries, select gifts and complete purchases. They need only basic skills in browsing items, selecting items and making secure online payments.

2.4: Constraints:

a. Time Constraint: The project must be completed and delivered within a 14-week timeframe.

b. Team Size: The development team is limited to 3 members, which impacts workload distribution and project management.

c. Weekly Feedback: Regular feedback must be incorporated, as weekly progress reports and reviews are required.

d. Technology Constraint: Java must be used as the primary programming language for the development of the software.

2.5: Assumptions and Dependencies:

a. Internet Access: It is assumed that user will have access to internet to interact with the GiftCart System.

b. User Registration: It is assumed that only register users will be able to create and manage registries.

c. Payment Gateway Availability: The system depends on the availability of third party payment gateway for processing secure transactions.

3. Specific Requirements

3.1. Functions

Use Case UC1: Registration

- **Primary Actor:** Admin (Shop Owner), Host, Guest
- **Stakeholders and Interest:**
 - Admin: wants to create account manage catalogue.
 - Host: wants to create account to manage gift registries for their event.
 - Guest: wants to create account to select gifts for their friend or family.
- **Preconditions:** None
- **Postconditions:** The user is successfully registered.
- **Input/Output:** Inputs are username, password, email.
Outputs are confirmation message or error message.
- **Main Success Scenario:**
 1. User enters username, email and selects role (Admin, Host, Guest).
 2. System calls **SendPintoEmail()** and sends a pin on email entered.
 3. Users enters the pin number.
 4. System calls **VerifyPin()** and verifies the pin number.
 5. System asks the user to enter password.
 6. Users enters desired password.
 7. System directs the user to specific dashboard.
- **Alternate Scenario:**
 - 2a. Invalid Email:**
 1. System shows invalid email message.
 2. System asks the user to enter email again.
 3. Users starts from step1 by only entering email.
 - 2b. Email Already Registered:**

Users enter email that is associated with an existing account with the same role.

 1. System displays email already in use.
 2. System prompts the users to login or reset password.
 - 4a. Incorrect Pin Number:**
 1. System notifies the users that pin is not matched.
 2. System displays option of send again.
 3. User click send again.
 4. Step 2 onward is repeated.

- **Frequency of Occurrences:** Can be frequently as new users are registered.
- **Special Requirements:**
 1. The system should send the pin number on email within 4 seconds to not keep the users waiting.
 2. System should verify the pin within 3 seconds.

Use Case UC2: Login

- **Primary Actor:** Admin (Shop Owner), Host, Guest
- **Stakeholders and Interest:**

Admin: wants to add /update/remove/view items in catalogue.

Host: wants to create event or add/update/remove/delete/view items from registry.

Guest: wants to select/view/remove/update gifts.
- **Preconditions:** Users must have a registered account.
- **Postconditions:** User has successfully logged in and has access to the dashboard.
- **Input/Output:**

Inputs are username, password.

Outputs are confirmation/success message, error message and access to catalogue/gift registry.
- **Main Success Scenario:**
 1. Users enter username and password into the login form.
 2. System calls **VerifyLoginCredentials()** and verifies the entered credential.
 3. System logs user in.
 4. System displays dashboard to the user.
- **Alternate Scenario:**
 - 1a. **Forgot Password:**
 1. User click on forgot password.
 2. System direct user to password recovery page.
 - 2a. **Invalid Credentials:**
 1. System displays error message and tells user to login again.
 2. User starts from step1.
- **Frequency of Occurrences:** Can be nearly continuous (high).
- **Special Requirements:**
 1. After 5 logins attempts system calls **LockUserAccount()** and the user account should be locked for 1hour to avoid security threats.

2. The system should verify the login credentials within 2 seconds to not keep the user waiting and ensure smooth user experience.
3. User account should be locked out automatically after 30 minute of inactivity.

UC3: Add items to Catalogue

- **Primary Actor:** Admin (Shop Owner).
- **Stakeholders and Offstage Actors:**

Host: interested in the successful creation of catalogue so he can create a gift registry for events.

Guests: want to buy items that have clear descriptions and good prices.

Developer Team: interested in successful working of their software.

- **Preconditions:** Admin must be logged into the system and may require user manual to properly use the system.
- **Post Conditions:** Item is displayed in the catalogue with necessary descriptions (name, price, product photos, and item stock information), item is given an ID and database is updated.
- **Input/Output:**

Items description like name, price, stock quantity, and photos are input to the system.

Confirmation/success messages, error messages, updated catalogue and database information are outputs.

- **Main Success Scenario:**
 1. Admin selects add new item button.
 2. The system displays a form to enter item details.
 3. Admin enters items with details such as name, price per unit, photos, stock count, description, and submits the form.
 4. The system adds the new item to the database and confirms success.
 5. The item is added to the catalogue and is displayed.
- **Alternate Scenarios:**
 - ***System Fails:**
 1. At any point in time the system fails due to some technical reasons.
 2. Admin restarts the system.
 3. The system requires rebooting.
 4. Admin selects reboot the system option.

5. The system rolls back any incomplete transactions and displays errors while adding items to the catalogue.
6. Admin starts with step 1.

3a. Admin forgets a mandatory description field:

1. The system displays errors in empty fields.
2. Admin enters the mandatory description field and resubmits the form.
3. The system continues to step 4.

4a. Item already exists in the catalogue:

1. The system prompts admin to cancel addition or update existing items.
2. Admin cancels addition and resumes with step 5.

2a. Admin updates the item:

1. Admin continues with *Update item in catalogue* use case.

- **Frequency of Occurrence:** Frequency may increase during peak days such as holidays, festivals and special days, as new arrivals in items and gifts are introduced. May be less frequent in normal days.
- **Special Requirements:**
 1. The system shall call **ValidateInputDetails()** and validate inputs.
 2. The system shall process addition of item within 2 seconds.

UC4: Update items in Catalogue

- **Primary Actor:** Admin (Shop Owner).
- **Stakeholders and Offstage Actors:**

Host, interested in the successful updating of catalogue so he can create a gift registry with updated and brand-new items.

Guests want to buy items that have clear and correct descriptions and good prices.

Developer Team interested in successful working of their software.

- **Preconditions:** Admin must be logged into the system; catalogue must have one or more items to update and may require a user manual to properly use the system.
- **Post Conditions:** Item is updated in the catalogue with new descriptions (name, price, product photos, and item stock information), and database is updated.
- **Input/Output:**

Items description like name, price, stock quantity, and photos are input to the system.

Update confirmation/success messages, error messages, updated catalogue and database information are outputs.

- **Main Success Scenario:**
 1. Admin selects an existing item by item ID in the search box.
 2. The system displays a form to enter new details for that selected item.
 3. Admin enters new/updated details such as name, price per unit, photos, stock count, description, and submits the form.
 4. The system adds the new updated item details to the database and confirms update success.
 5. The updated item is added to the catalogue and is displayed.

- **Alternate Scenarios:**

- ***System Fails:**

- 1. At any point in time the system fails due to some technical reasons.
 2. Admin restarts the system.
 3. The system requires rebooting.
 4. Admin selects reboot the system option.
 5. The system rolls back any incomplete transactions and displays errors while adding items to the catalogue.
 6. Admin starts with step 1.

- **1a. Item not found:**

- 1. Admin enters an invalid or non-existent item ID
 2. The system displays an error message ("Item not found") and prompts the admin to search again or check the item ID.
 3. Admin re-enters the ID and re-submits the search.
 4. The system continues to step 2.

- **3a. Admin forgets a mandatory description field:**

- 1. The system displays errors in empty fields.
 2. Admin enters the mandatory description field and resubmits the form.
 3. The system continues to step 4.

- **Frequency of Occurrence:** Frequency may increase during peak days such as holidays, festivals and special days, as new arrivals in items and gifts are introduced and special discounts in prices and time-specific deals. May be less frequent in normal days.

- **Special Requirements:**

1. The system shall call **ValidateInputDetails()** and validate inputs.
2. The system should reflect updated items in the catalogue within two seconds to ensure minimal delay.

UC5: Delete items from Catalogue

Primary Actor: Admin (Shop Owner).

Stakeholders and Offstage Actors:

1. **Host**, interested in the successful updating of catalogue so he can create a gift registry with updated and brand-new items.
2. **Guests** want to buy items that have clear and correct descriptions and good prices.
3. **Developer Team** interested in successful working of their software.

Preconditions: Admin must be logged into the system; catalogue must have one or more items to delete and may require a user manual to properly use the system.

Post Conditions: Item is deleted from the catalogue, and catalogue is updated.

Input/Output:

1. Delete confirmation/success messages, error messages, updated catalogue information are outputs.

Main Success Scenario:

1. Admin selects an existing item by item ID in the search box.
2. The system displays a form to enter new details for that selected item.
3. Admin selects delete item option.
4. The system calls **DeleteItemFromCatalogue()** and deletes the item from the catalogue.
5. The updated catalogue is displayed.

Alternate Scenarios:

***System Fails:**

At any point in time the system fails due to some technical reasons.

1. Admin restarts the system.

2. The system requires rebooting.
3. Admin selects reboot the system option.
4. The system rolls back any incomplete transactions and displays errors while adding items to the catalogue.
5. Admin starts with step 1.

1a. Item not found:

Admin enters an invalid or non-existent item ID

1. The system displays an error message ("Item not found") and prompts the admin to search again or check the item ID.
2. Admin re-enters the ID and re-submits the search.
3. The system continues to step 2.

Frequency of Occurrence: Frequency may increase during peak days such as holidays, festivals and special days, as new arrivals in items and gifts are introduced and special discounts in prices and time-specific deals. May be less frequent in normal days.

Special Requirements:

1. The system should reflect updated items in the catalogue within two seconds to ensure minimal delay.

UC6: View Catalogue

Primary Actor: Admin (Shop Owner).

Stakeholders and Offstage Actors:

1. **Host**, interested in the successful maintenance of catalogue so he can create a gift registry with updated and brand-new items.
2. **Guests** want to buy items that have clear and correct descriptions and good prices.
3. **Developer Team** interested in successful working of their software.

Preconditions: Admin must be logged into the system; and may require a user manual to properly use the system.

Post Conditions: Items are displayed from the catalogue.

Input/Output:

2. Items lists are output.

Main Success Scenario:

1. Admin selects view catalogue option.
2. The system displays catalogues and items.

Alternate Scenarios:***System Fails:**

At any point in time the system fails due to some technical reasons.

1. Admin restarts the system.
2. The system requires rebooting.
3. Admin selects reboot the system option.
4. The system rolls back any incomplete transactions and displays errors while adding items to the catalogue.
5. Admin starts with step 1.

Frequency of Occurrence: May be variable.

Special Requirements:

1. The system should display catalogue within 2 seconds of display request.

UC7: Create Event

- **Primary Actor:** Host (Registered User).
- **Stakeholders and Offstage Actors:**

Admin, interested in the successful creation of events so he can earn profit and sales.

Guests want to buy gifts for events.

Developer Team interested in successful working of their software.

- **Preconditions:** Host must be logged into the system; may require a user manual to properly use the system.
- **Post Conditions:** Event is created with event details like event name, event description, event date, event deadline (for gifts) and host name.
- **Input/Output:**

Event details like name, description, date, deadline and host name are input into the system.

Event creation confirmation/success messages, error messages are outputs.

- **Main Success Scenario:**

1. Host selects the “Create Event” button.
2. The system displays a form to enter details for events to be created.
3. Host enters details such as event name, description, event date, deadline for event gifts, host name and submits the form.
4. The system adds the event details to the database and confirms event creation with an empty gift registry.
5. The event is displayed on the Host profile with empty gift registry and Host may continue with **Add Item to Registry** use case

- **Alternate Scenarios:**

- ***System Fails:**

1. At any point in time the system fails due to some technical reasons.
2. The system displays error message “Please try again after some time”.
3. Host restarts the system.
4. The system restores and rolls back any incomplete transactions.
5. The Host starts with step 1 again.

- **3a. Missing required information:**

1. The system displays an error message ("Fill required information") and prompts the host to re-submit again.
2. The host enters necessary information and re-submits.
3. The system continues to step 4.

- **3b. Duplicate Events:**

1. The system displays errors “Event with same name and date already exists” and prompts the host to continue or review existing events.
2. Host cancel the current event creation:
3. System displays message “Event cancelled”.
4. Host continues with current events:
5. System assigns a post-fix to the event e.g. Birthday().
6. The system continues to step 4.

- **Frequency of Occurrence:** Frequency may increase during peak days such as holidays, festivals and special days such as wedding seasons or birthdays.

- **Special Requirements:**

1. The system shall call **ValidateInputDetails()** and validate inputs.
2. The system should create events within 2 seconds from submission to confirmation.
3. Hosts cannot create an event which has an event date more the 60 days after the event creation, or when the event is less than 4 days gaped from the day of event creation.

UC8: Add Item to Registry

- **Primary Actor:** Host (Registered User)
- **Stakeholders And Offstage Actors:**

Guest: Interested in viewing and selecting items from the list as gifts for the event.

Shop Owner: Wants items from catalogue to be included in list.

System Administrator: Ensures the system runs smoothly.

Development Team: Interested in accurate performance of feature.

- **Preconditions:**
 1. Host is already logged to system.
 2. The shop`s catalogue is up to date with available items.
- **Input/Output:**

Input: Items details such as id and quantity.

Output: Success messages, error messages, updated registry view, and confirmation of item added with selected quantities.

- **Main Success Scenario:**
 1. The host selects the option to add items to registry.
 2. The system displays available items along with their descriptions (name,price,stock,quantity).
 3. The host browses items from catalogue.
 4. The host selects one or more items along with their desired quantities.
 5. The host confirms the selection.
 6. The system adds selected items to gift registry and update it in real times.
 7. A conformation message is displayed, and registry is updated for the guest to view
- **Postconditions:**

1. The selected items are successfully added to the registry with their respective quantities.
 2. Database is updated with hosts selection and system tracks availability based on chosen quantities.
- **Alternate Scenarios:**
 - 2a: Host forgets to select a quantity or enters an invalid value:**
 1. The system alerts the host with error message.
 2. The host review the error message and re-submit.
 3. The system proceeds next.
 - 2b: The system fails due to technical issues:**
 1. The system notifies the host of technical issues and logs the error.
 2. The system rolls back any incomplete transaction to make sure data consistency.
 3. The host can retry of adding items to registry.
 4. The system continues to the point of failure.
 - 4a: Item is out of stock or requested quantity exceeds limit.**
 1. The system displays an error message indicating insufficient stock.
 2. The host can either
 3. Reduce the quantity.
 4. Chose a different item.
 - 6a: Duplicate items already exist in list:**
 1. If item being added already exist in the registry, the system prompts the host to cancel addition.
 2. The system proceeds accordingly and updates registry.
- **Frequency of Occurrence:** Occur frequently especially for popular occasions like Weddings, birthdays, and other special events. The frequency may rise during special promotions and less in normal days.
 - **Special Requirements:**
 1. The system shall call **ValidateInputDetails()** and validate inputs.
 2. The system shall process addition of item within 2 seconds.

UC9: Update Item in Registry

- **Primary Actor:** Host (Registered User)
- **Stakeholders And Offstage Actors:**

Guest: Interested in viewing and selecting items from the list as gifts for the event.

Shop Owner: Wants items from catalogue to be included in list.

System Administrator: Ensures the system runs smoothly.

Development Team: Interested in accurate performance of feature.

- **Preconditions:**
 1. Host is already logged to system.
 2. The shop`s catalogue is up to date with available items
- **Main Success Scenario:**
 1. The host selects the option to update items in registry.
 2. The system displays available items along with their descriptions (name,price,quantity).
 3. The host select an item for update.
 4. The host modifies the quantity, adds a note or changes other attributes if available.
 5. The host confirms the update.
 6. The system applies the updates in real time adjusting the registry accordingly.
 7. A conformation message is displayed by the system, and registry is refreshed for the host.
- **Postconditions:**
 1. The selected items in the registry will reflects the changes made by host.
- **Alternate Scenarios:**

The system fails due to technical issues:

 2. The system notifies the host of technical issues and logs the error.
 3. The host should restart the system.
 4. The system rolls back any incomplete transaction to make sure data consistency.
 5. The host can retry of updating any item to registry.

4a: Invalid Input or missing information:

1. If the host fails to specify valid quantity, adds invalid note, or leaves required fields empty, the system should display the error message with specific information.
 2. The host review the error message and corrects the information.
 3. Once corrected the system validates the new input and proceeds with the update.
- **Frequency of Occurrence:** Occur frequently especially for popular occasions like Weddings, birthdays, and other special events. The frequency may rise during special promotions and less in normal days.
 - **Special Requirements:**
 1. The system shall call **ValidateInputDetails()** and validate inputs.
 2. The registry should reflect the updates within two seconds.

UC10: Delete item from registry:

- **Primary Actor:** Host (Registered User)
- **Stakeholders And Offstage Actors:**
 - Guest:** Interested in viewing and selecting items from the list as gifts for the event.
 - Shop Owner:** Wants items from catalogue to be included in list.
 - System Administrator:** Ensures the system runs smoothly.
 - Development Team:** Interested in accurate performance of feature.
- **Preconditions:**
 1. Host is already logged to system.
 2. The shop`s catalogue is up to date with available items
- **Main Success Scenario:**
 1. The host selects the option to delete items from registry.
 2. The system displays available items along with their descriptions (name,price,quantity).
 3. The host select one or more items for delete.
 4. The host confirms deletion of items.
 5. The system removes the selected items from registry in real time.
 6. A confirmation message is displayed by the system and registry is refreshed for host and guest.
- **Postconditions:**

1. The selected items will be permanently removed from the registry by the system.
- **Alternate Scenarios:**
 - The system fails due to technical issues:**
 2. The system notifies the host of technical issues and logs the error.
 3. The system rolls back any incomplete transaction to make sure data consistency.
 4. The host can retry of updating any item to registry.
 - 4a: Missing confirmation:**
 1. If the host does not confirm the deletion (e.g., accidentally navigates away), the system alerts the host and cancels the operation.
 2. The host can reselect the items for deletion and confirm to proceed.
 - 6a: Multiple users viewing the registry simultaneously.**
 1. If the guest is viewing the registry while the host removes an item from registry the system updates the guest's view to reflect the current registry state.
 2. Any selected item must not reappear for the guest if it was successfully deleted by host.
- **Frequency of Occurrence:** Occur frequently especially for popular occasions like Weddings, birthdays, and other special events. The frequency may rise during special promotions and less in normal days.
- **Special Requirements:**
 1. The registry should reflect the deletion within two seconds.
 2. A confirmation prompt should be displayed before finalizing any deletion to avoid accidental deletion of items.
 3. Only the host has permission to delete items from registry.
 4. The system should support the multiple user viewing the registry simultaneously adjusting the view for each user if deletions are occur during browsing.

UC11: View Registry

Primary Actor: Host (Registered User), Guest(Friend)

Stakeholders And Offstage Actors:

1. **Guest:** Interested in viewing and selecting items from the list as gifts for the event.
2. **Shop Owner:** Wants items from catalogue to be included in list.
3. **System Administrator:** Ensures the system runs smoothly.
4. **Development Team:** Interested in accurate performance of feature.

Preconditions:

- User is already logged to system.

- The shop`s catalogue is up to date with available items

Main Success Scenario:

1. The user selects the option to view the registry.
2. The system displays the registry with all available items along with their descriptions (name,price,quantity).
3. The user scrolls through or searches within the registry to view specific items.
4. The user can filter items based on attributes (e.g., category or price).
5. The system updates the display dynamically based on host action such as sorting or filtering.
6. The user can complete the viewing and exits the registry.

Postconditions:

- The registry remains unchanged by this action, only viewed by the user.

Alternate Scenarios:*** The system fails due to technical issues:**

1. The system notifies the host of technical issues and logs the error.
2. The host should restart the system.
3. The system rolls back any incomplete transaction to make sure data consistency.
4. The host can retry of updating any item to registry.

2a: No item in the registry:

1. If the registry is empty the system displays a message that no item is currently available in the registry.

Frequency of Occurrence:

Occur frequently especially for popular occasions like Weddings, birthdays, and other special events. The frequency may rise during special promotions and less in normal days.

Special Requirements:

1. Any sorting or filtering apply by the user on items search should apply within two seconds.
2. The system should allow smooth navigation within the registry without any delay.
3. Input fields such as search, and filters must include validation checks.

Use case UC12: Select Gift

- **Primary Actor:** Guest
- **Stakeholders and Interest:**

Guest: wants selection of gift based on his preferences and budget.

Host: wants to receive gifts according to his/her needs and preferences.

Shop owners want to offer a variety of gifts to satisfy guest needs and preferences. Wants to offer gifts at different prices according to host financial capabilities.

- **Preconditions:** Guest must be logged in before selecting gifts.
- **Postconditions:** Gifts are selected, running total and net total is calculated and displayed.
- **Main Success Scenario:**
 1. Guests select the event.
 2. The system displays gift lists with description (name, price, quantity limit) of that event.
 3. Guests browse the list and click to select items, with quantity.
 4. The system marks the selected items.
 5. System calls **CalculateRunningTotal()** and **CalculateNetTotal()** and displays running total and net total respectively.
- **Alternate Scenario:**
 - System Fails:**
 1. At any instant the system freezes or crashes
 2. Guests refresh the web.
 3. The system returns to normal state and proceeds to step 3.
 - 2a. The System restarts:**
 1. The guest again selects gifts and proceeds.
 - 1a. Deadline exceeds:**
 2. Deadline reaches mid selection or before selection.
 3. The system displays deadlines exceeding message.
 4. The guest cancels selection.
 - 3a. Item sold out:**
 1. Items have been sold out.
 2. The system displays messages showing items that are sold out.
 3. The guest selects some other item
- **Frequency of occurrences:** Vary during normal days but very high during holidays, festivals and celebration days.
- **Special Requirements:**
 1. A confirmation prompt should be displayed before finalizing any selection of gift.

Use Case UC13: Update Gift

- **Primary Actor:** Guest
- **Stakeholders and interest:**
Guest: wants to modify the quantity of a gift he/she has selected.
- **Preconditions:** Guest must have selected a gift already
- **Postconditions:** Quantity of selected gift is updated.
- **Main Success Scenario:**
 1. Guest clicks on gift which he/she has selected.
 2. System displays gift details including description, piece, quantity selected.
 3. Guest clicks increment/decrement quantity.
 4. System increment/decrement the gift quantity.
- **Alternate Scenario:**
System Fails: At any point in time the system fails due to some technical reasons.
 1. The system displays error message “Please try again after some time”.
 2. Host restarts the system.
 3. The system restores and rolls back any incomplete transactions.
 4. The Host starts with step 1 again.
- **4a. cannot Increment:**
System cannot increment the quantity of selected gift as require quantity has reached.
 1. System displays cannot increment the quantity of selected gift as require quantity has reached message.
- **Frequency of Occurrences:** can be frequently.
- **Special Requirements:**
 1. System shall update the quantity in real time.

UC14: Remove Gift:

Primary Actor: Guest (Friend)

Stakeholders And Offstage Actors:

5. **Guest:** Interested in viewing and selecting items from the list as gifts for the event.
6. **Shop Owner:** Wants items from catalogue to be included in list.
7. **System Administrator:** Ensures the system runs smoothly.
8. **Development Team:** Interested in accurate performance of feature.

Preconditions:

- Guest is already logged to system.
- The shop's catalogue is up to date with available items

Main Success Scenario:

7. The Guest selects the gifts from gift list marked as reserved.
8. The Guest selects remove option to remove selected gift or gifts.
9. The Guest confirms the removal of selected gifts from their item list.
10. The system removes the item or items from the guest's list, making it available for other guests if not yet purchased.
11. A confirmation message is displayed by the system and registry is refreshed for the guest.

Postconditions:

- The selected items will be permanently removed from the item list by the system.

Alternate Scenarios:*** The system fails due to technical issues:**

5. The system notifies the host of technical issues and logs the error.
6. The system rolls back any incomplete transaction to make sure data consistency.
7. The host can retry of updating any item to registry.

3a: Guest does not confirm removal of gift.

1. If the guest does not confirm the removal, the system alerts them and cancels the action.
2. The guest may reselect the item for removal if they wish to proceed later.

Frequency of Occurrence:

Occur frequently especially for popular occasions like Weddings, birthdays, and other special events. The frequency may rise during special promotions and less in normal days.

Special Requirements:

- The registry should reflect the removal within two seconds.
- A confirmation prompt should be displayed before finalizing any removal to avoid accidental removal of items.
- Only the guest can remove their reserved items without impacting other users' selections or the host's registry list.

- The system should support the multiple user viewing the registry simultaneously adjusting the view for each user if removals are occur during browsing.

Use Case UC15: View Gifts

- **Primary Actor:** Host, Guest
- **Stakeholders and Interest:**
Host: wants to know about how many gifts have been selected from his/her list.
Guest: wants to view gifts from his/her friend registry.
- **Preconditions:** The user must be logged in.
- **Postconditions:** The user has browsed the gift registry.
- **Input/Output:**
Input is only selection of gift (clicks).
Output is Gift list, Gift details.
- **Main Success Scenario:**
 1. Users selects registry.
 2. System displays gift list.
 3. User browses the list.
 4. Users selects a gift to view more detail.
 5. System shows information about gift selected (Description, Price, Quantity Available, Image).
- **Alternate Scenario:**
2a. No Gifts Available:
 1. System displays no gifts are currently available.
- **Frequency of Occurrences:** Can be nearly continuous.
- **Special Requirements:**
 1. The system must provide real time updates on item availability.

Use case UC16: Make Payment

- **Primary Actor:** Guest
- **Stakeholders and interest:**
Guest: wants secure and seamless payment process so that he can continue with the purchase of gift.
Admin (shop owner): wants to receive payment for item he sold.
- **Input/Output:**
Payment information including card number, expiry date, CVC are input.
Payment confirmation, error message are output.
- **Preconditions:** The guest must have selected one or more gift for purchase
- **Postconditions:** The payment is processed successfully and order is confirmed.
- **Main Success Scenario:**

1. The guest select proceed to payment option.
 2. System displays total and a form to enter payment information including card number, CVC, expiry date.
 3. Guest enters the required payment information.
 4. System calls **ValidatePaymentInformation()** and verify the payment info.
 5. System process payment and show Payment Successful message.
- **Alternate Scenario:**
System Fails: At any point in time the system fails due to some technical reasons.
 1. The system displays error message “Please try again after some time”.
 2. Host restarts the system.
 3. The system restores and rolls back any incomplete transactions.
 4. The Host starts with step 1 again.
- 4a. Invalid payment information:**
1. System shows invalid information.
 2. System show option of enter detail again.
 3. Guest clicks enter detail again option proceed with step2
- **Frequency of Occurrences:** can be nearly continuous.
 - **Special Requirements:**
 1. The system shall validate payment information in real time.

3.2. System Functions

1. **Backup:**
 - **Preconditions:** Availability of user’s data (Username, password, transactions performed), item information, registry details for backup.
 - **Postconditions:** A detail of user data (Username, password, transactions performed), item information, and registry detail is created and stored securely.
 - **Input Source:** User data, item information, registry details.
 - **Output:** Successful backup file.
 - **Actions:**
 1. System collects relevant user data, item information, registry detail.
 2. System writes collected data to a backup file.
 3. System store backup data in a designated secure location
- **Special Requirements:**
 1. The system shall perform backup at schedule time interval.
 - 2.

3.3. Sub Functions:

1. SendPintoEmail():

- **Preconditions:** User has entered valid email address.
- **Postconditions:** A pin is sent to user email.
- **Input Source:** Users email address.
- **Output:** Confirmation message that a pin has been sent.
- **Actions:**
 1. System generates pin.
 2. System sends pin to provided email address.
- **Special Requirements:**
 1. The system should generate and send PIN number within 4 seconds

2. VerifyPin():

- **Preconditions:** User has received PIN.
- **Postconditions:** System confirms whether entered PIN is valid.
- **Input Source:** User entered PIN.
- **Output:** Confirmation that PIN number is valid.
- **Actions:**
 1. System compares entered PIN with generated PIN.
- **Special Requirements:**
 1. System should verify the PIN within 3 seconds.

3. LockUserAccount():

- **Preconditions:** Five failed login attempts are detected.
- **Postconditions:** User account is locked.
- **Input Source:** Login attempts log.
- **Output:** Account lock notification.
- **Actions:**
 1. System Lock user account and set timer for 1 hour.
- **Special Requirements:**
 1. Not applicable.

4. VerifyLoginCredentials():

- **Preconditions:** User has a registered account.
- **Postconditions:** User has logged in and has access to dashboard.

- **Input Source:** Username and password.
- **Output:** Confirmation of successful login.
- **Actions:**
 1. System checks entered credentials against stored data.
- **Special Requirements:**
 1. The system should verify the login credentials within 2 seconds.

5. DeleteItemFromCatalogue():

- **Preconditions:** Item to be deleted must exist in the catalogue.
- **Postconditions:** Item is deleted from catalogue.
- **Input Source:** Item ID of the item to be deleted.
- **Output:** Confirmation message that item has been successfully deleted from catalogue.
- **Actions:**
 1. System verifies existence of the item using provided item ID.
 2. System deletes the item from catalogue
- **Special Requirements:**
 1. The system should complete the deletion process within 2 seconds

6. ValidateInputDetails():

- **Preconditions:** Admin has logged in and has entered input details.
- **Postconditions:** Input details are validated.
- **Input Source:** Input details form.
- **Output:** Confirmation of valid input.
- **Actions:**
 1. System checks that mandatory fields are filled out correctly.
- **Special Requirements:**
 1. The system shall validate input fields within 3 seconds

7. CalculateRunningTotal()

- **Preconditions:** Guest has selected one or more gift along with quantity.
- **Postconditions:** System displays updated running total.
- **Input Source:** Selected gift and their quantities.
- **Output:** System displays running total to guest.
- **Actions:**

1. System multiplies each gift price by its quantity to get calculated amount.
 2. System sums all calculated amount to get running total
- **Special Requirements:**
 1. The system shall complete calculation in under 2 seconds.

8. CalculateNetTotal():

- **Preconditions:** The running total has been calculated.
- **Postconditions:** System displays updated net total.
- **Input Source:** Running total amount and GST.
- **Output:** System displays net total to guest.
- **Actions:**
 1. System apply GST on running total to get net total.
- **Special Requirements:**
 2. The system shall complete calculation in under 2 seconds.

9. ValidatePaymentInformation():

- **Preconditions:** Host has initiated payment process.
- **Postconditions:** System has validated payment information.
- **Input Source:** Payment details (card number, expiration date, CVV)
- **Output:** Confirmation of valid information input.
- **Actions:**
 1. System checks format of credit card number.
 2. System verifies that expiration date is in future.
 3. System checks format of CVV.
 4. System can also perform validation check against payment gateway.
- **Special Requirements:**
 1. The system shall perform validation in real time.

3.4. Performance Requirement:

1. The system should send the pin number on email within 4 seconds to not keep the users waiting.
2. System should verify the pin within 3 seconds.
3. After 5 logins attempts the user account should be locked for 1hour to avoid security threats.
4. The system should verify the login credentials within 2 seconds to not keep the user waiting and ensure smooth user experience.
5. User account should be locked out automatically after 30 minutes of inactivity.
6. The system must provide real time updates on item availability.
7. The registry/catalogue should reflect the updates within two seconds.
8. Input fields such as quantity and item attributes must include validation checks.
9. The registry should reflect the deletion within two seconds.
10. A confirmation prompt should be displayed before finalizing any deletion to avoid accidental deletion of items.
11. Only the host and admin have permission to delete items from registry and catalogue respectively.
12. The system should support the multiple users viewing the registry simultaneously adjusting the view for each user if deletions are occurred during browsing.
13. The system should create events within 2 seconds from submission to confirmation.
14. Hosts cannot create an event which has an event date more the 60 days after the event creation, or when the event is less than 4 days gaped from the day of event creation.
15. The system should reflect updated items in the catalogue within two seconds to ensure minimal delay.
16. The system shall process addition of item within 2 seconds.
17. Any sorting or filtering apply by the user on items search should apply within two seconds.
18. The system should allow smooth navigation within the registry without any delay.
19. Input fields such as search, and filters must include validation checks.
20. System shall update the quantity in real time.
21. The registry should reflect the removal within two seconds.
22. A confirmation prompt should be displayed before finalizing any removal to avoid accidental removal of items.
23. Only the guest can remove their reserved items without impacting other users' selections or the host's registry list.
24. The system should support the multiple users viewing the registry simultaneously adjusting the view for each user if removals are occurred during browsing.
25. The system shall validate payment information in real time.

4. Appendix

4.1. Data Dictionary

Attribute Name	Description	Alias	Type	Length/Conditions
adminID	Unique identifier for each administrator		STRING	Length: 10, Must be unique
amount	Amount for payment		DECIMAL(10,2)	Must be positive
address	Address of guest and host		VARCHAR	Length: 255, Cannot be null
catalogueID	Unique identifier for each catalogue		STRING	Length: 10, Must be unique
catalogueTitle	Title of the catalogue		VARCHAR	Length: 30, Cannot be null
contactNo	Contact number of guest, host, admin, or shop		VARCHAR	Length: 15
dateCreated	Date when catalogue or registry was created		DATETIME	
deadline	Deadline for the event		DATETIME	
description	Detailed description of the catalogue		TEXT	Cannot be null
email	Email address of admin, guest, host, and shop		VARCHAR	Length: 40, Must be valid
eventDate	Date of the event		DATETIME	
eventDescription	Description of the event		TEXT	Cannot be null
eventID	Unique identifier for each event		STRING	Length: 10, Must be unique
eventName	Name of the event		VARCHAR	Length: 30, Cannot be null
giftID	Unique identifier for each gift		STRING	Length: 10

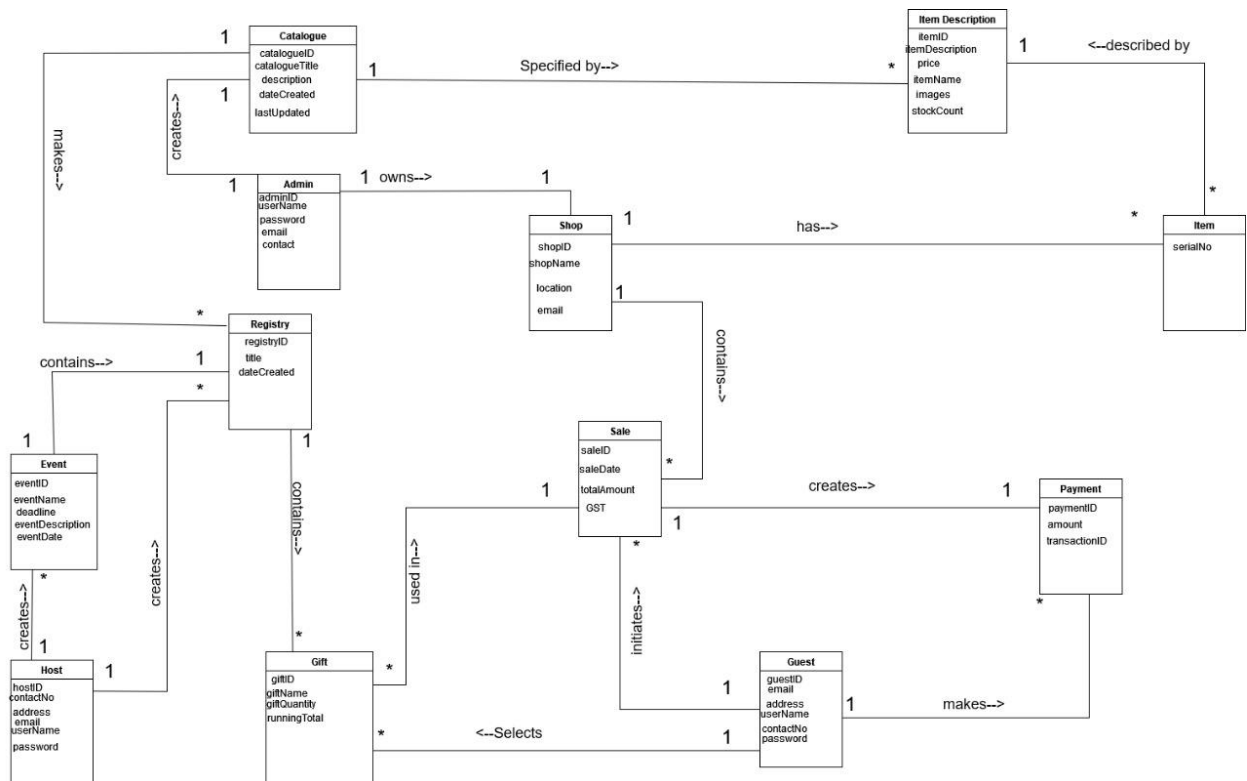
giftList	List of gifts selected	VARCHAR	
giftName	Name of the gift	VARCHAR	Length: 30, Cannot be null
giftQuantity	Quantity of the gift	FLOAT	Length: 3, Must be >= 0
GST	Taxes on sale	FLOAT	
itemDescription	Detailed description of the item	TEXT	Cannot be null
itemID	Unique identifier for each item description	STRING	Length: 10, Must be unique
itemList	List of items in registry	VARCHAR	
itemName	Name of the item	VARCHAR	Length: 30, Cannot be null
lastUpdated	Date of last update to catalogue	DATETIME	
location	Physical address of shop	VARCHAR	Length: 255
password	Password used for login of admin, host, guest	VARCHAR	Length: 255, Minimum 8 characters
paymentID	Unique identifier for each payment	STRING	Length: 10, Must be unique
price	Price of the item	DECIMAL(10,2)	Must be positive
registryID	Unique identifier for each registry	STRING	Length: 10, Must be unique
runningTotal	Running total of the gift	DECIMAL(10,2)	Must be positive
saleDate	Date on which sale was done	DATETIME	
saleID	Unique identifier for each sale	STRING	Length: 10, Must be unique
serialNo	Unique identifier for each item	STRING	Length: 10, Must be unique
shopID	Unique identifier for each shop	STRING	Length: 10, Must be unique
shopName	Name of the shop	VARCHAR	Length: 255, Cannot be null

stockCount	Stock count of the item	INT	Must be ≥ 0
title	Title of the registry	VARCHAR	Length: 255, Cannot be null
totalAmount	Total amount after sale	DECIMAL(10,2)	Must be positive
transactionID	Transaction ID of payment	STRING	Length: 10, Must be unique
userName	Username used for login of admin, host, guest	STRING	Length: 30, Cannot be null

Domain Objects in Dictionary

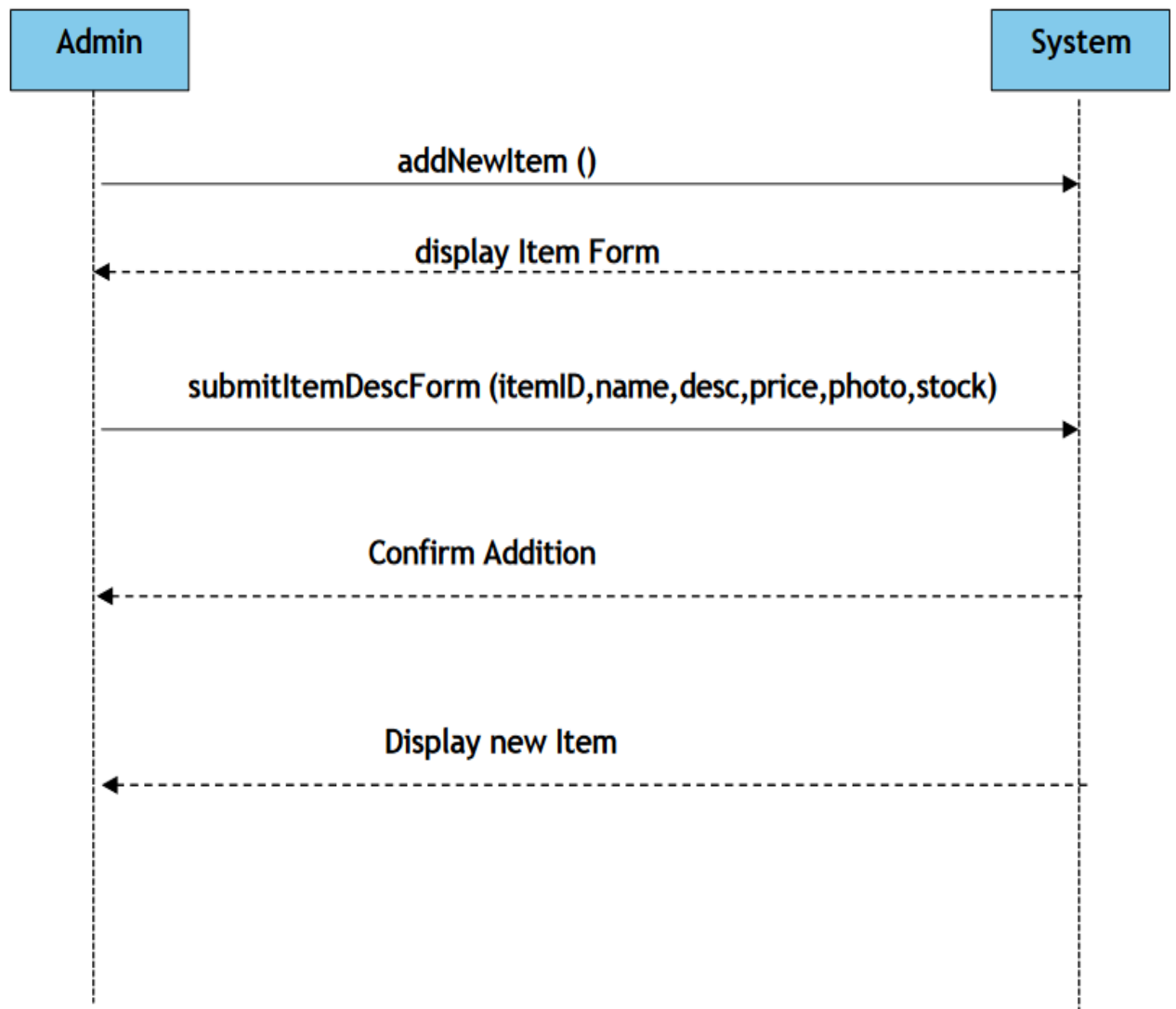
Object Name	Description	Attributes
Admin	System administrator who manages catalogue	adminID, userName, password, email, contactNo
Catalogue	List of products	catalogueID, catalogueTitle, description, dateCreated, lastUpdated
Event	Event created by host	eventID, eventName, deadline, eventDescription, hostName, eventDate
Gift	Gifts selected by guests from registry	giftID, giftName, giftQuantity, itemList
Guest	Guests purchase gifts from registry for hosts	guestID, email, address, userName, contactNo, password
Host	Host who creates and manages registry	hostID, contactNo, address, email, userName, password
Item	Items in catalogue	serialNo, itemDescription, price, itemName, images, stockCount
Payment	Payment made by host after purchase	paymentID, amount, transactionID
Registry	Registry created by host containing item list	registryID, title, dateCreated, itemList
Sale	Sale initiated by guest	saleID, saleDate, totalAmount, giftList, GST
Shop	Admin as the shop owner	shopID, shopName, location, email

4.2 Domain Diagram

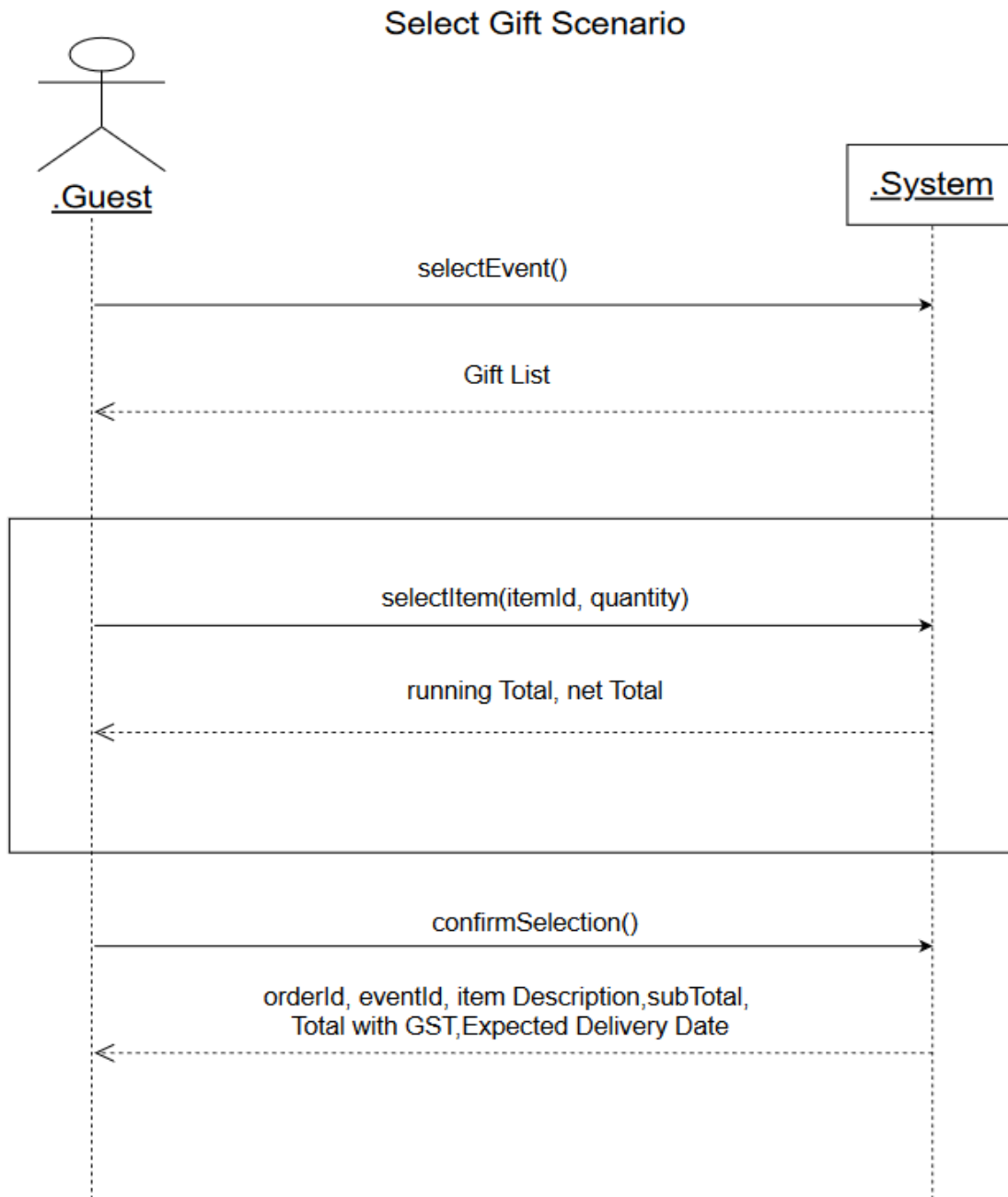


4.3 System sequence diagrams

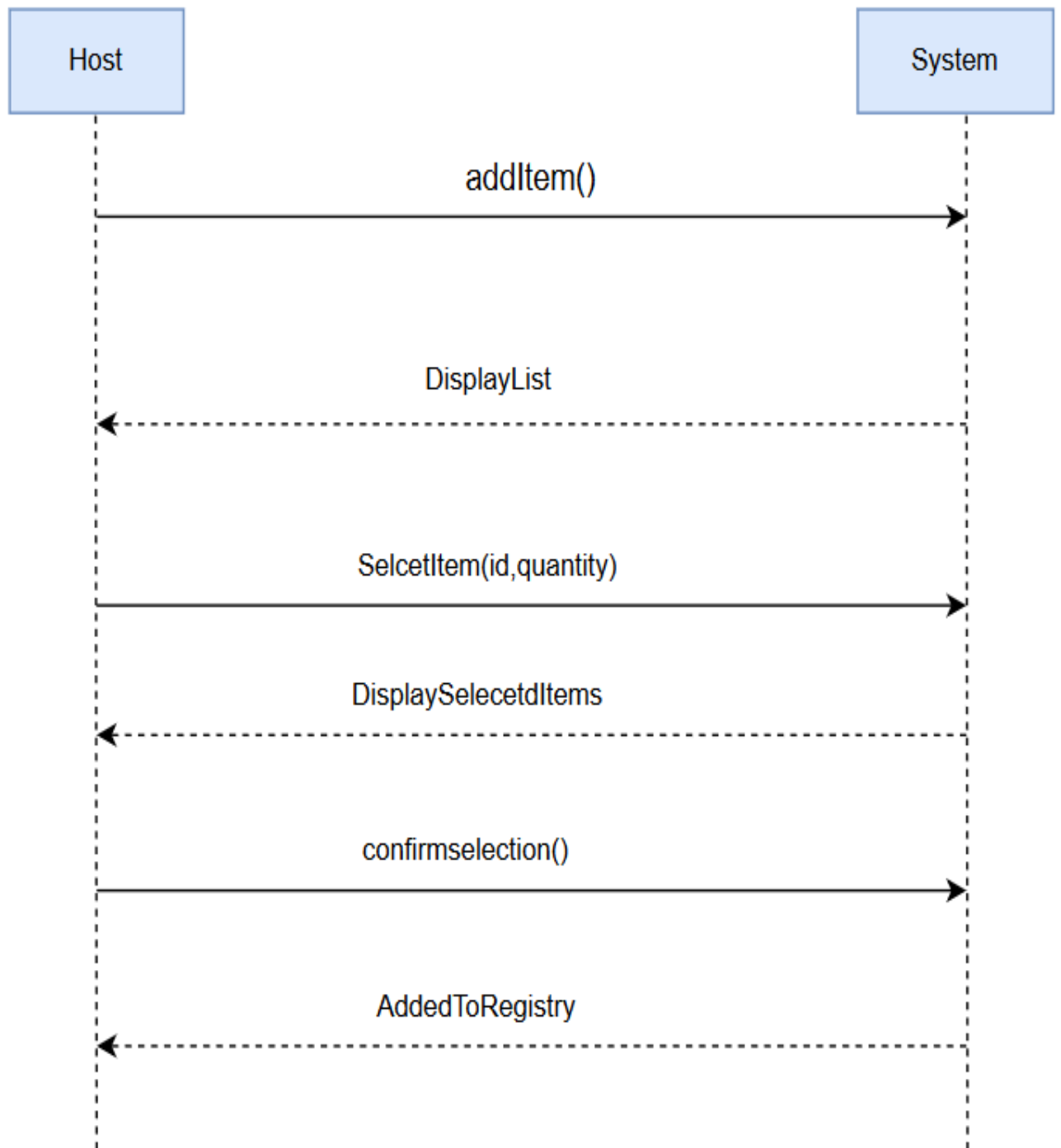
4.3.1. System sequence diagram for UC3



4.3.2. System sequence diagram for UC12



4.3.3. System sequence diagram for UC6



4.4. Use Case Diagram

